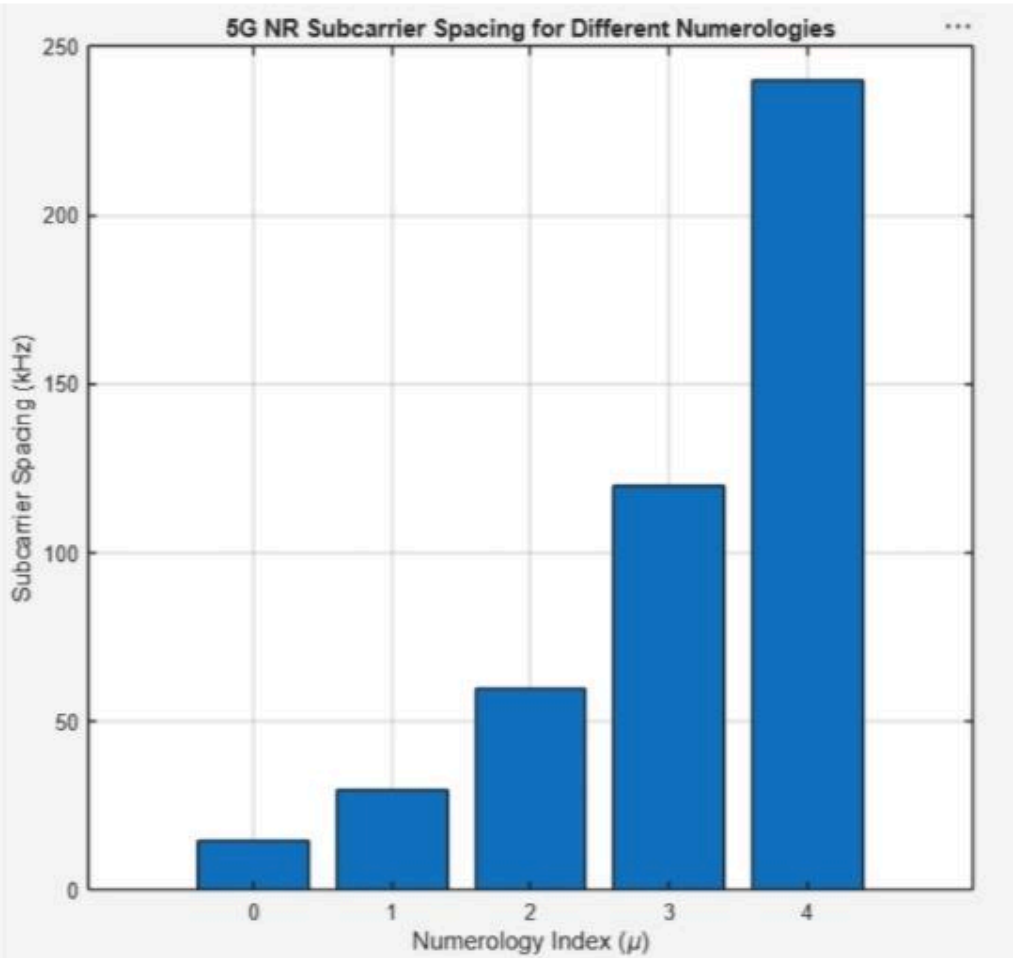


Ex.No.6: 5G NR Frame Structure Analysis with Different Subcarrier Spacing

Objective 1: Matlab Code and Output

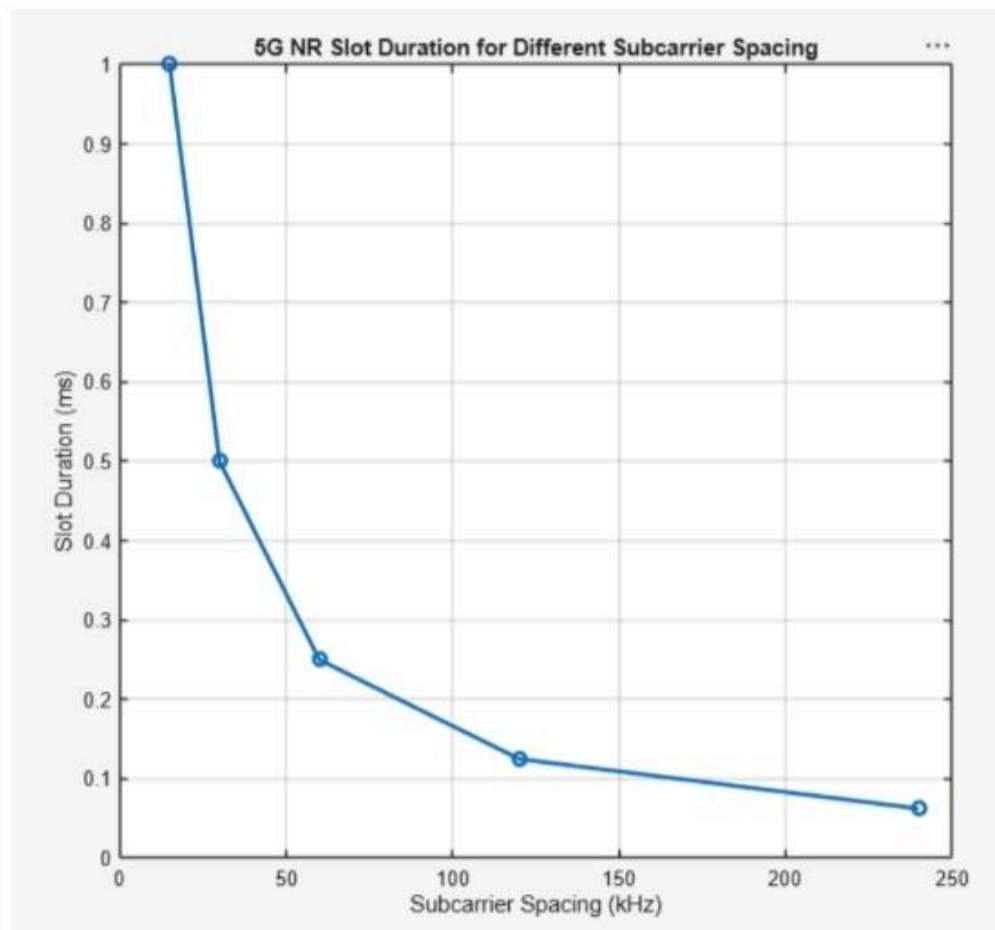
/MATLAB Drive/slot6.m

```
1   clc;
2   clear;
3   close all;
4   scs = [15 30 60 120 240];
5   mu = 0:4;
6   bar(scs)
7   grid on
8   set(gca,'XTick',1:5)
9   set(gca,'XTickLabel',mu)
10  % Subcarrier Spacing (kHz)
11  % Numerology Index
12  xlabel('Numerology Index (\mu)')
13  ylabel('Subcarrier Spacing (kHz)')
14  title('5G NR Subcarrier Spacing for Different Numerologies')
15
```



Objective 2: Matlab Code and Output

```
16   scs = [15 30 60 120 240];  
17   % Subcarrier Spacing (kHz)  
18   slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)  
19   plot(scs, slot, 'o-', 'LineWidth', 2)  
20   grid on  
21   xlabel('Subcarrier Spacing (kHz)')  
22   ylabel('Slot Duration (ms)')  
23   title('5G NR Slot Duration for Different Subcarrier Spacing')  
24
```



Objective 3: Matlab Code and Output

```
25     scs = [15 30 60 120 240];  
26     symbols = [14 14 14 14 14];  
27     figure  
28     bar(scs, symbols)  
29     % Subcarrier Spacing (kHz)  
30     % OFDM Symbols per Slot  
31     grid on  
32     xlabel('Subcarrier Spacing (kHz)')  
33     ylabel('Number of OFDM Symbols')  
34     title('OFDM Symbols per Slot for Different Subcarrier Spacing')
```

